1.40 2D Matrices: Combining Transformations

Worked Solutions

1 a)

BA.

1 b)

AC.

1 c)

CB.

1 d)

BCA.

1 e)

CAB.

2 a)

$$\mathbf{BA} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & -3 \\ 8 & 6 \end{pmatrix} = \begin{pmatrix} 8 & 6 \\ 2 & -3 \end{pmatrix}$$

2 b)

$$\mathbf{AB} = \begin{pmatrix} 2 & -3 \\ 8 & 6 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -3 & 2 \\ 6 & 8 \end{pmatrix}$$

3)

Reflection in the x-axis, $\mathbf{M} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Reflection in the y-axis, $\mathbf{N} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

$$\mathbf{NM} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

This is a rotation about the origin by 180°.

4 a)

Reflection in the line $y = x$.

4 b)

$\mathbf{A}^2 = \mathbf{I}$, $\mathbf{A}^4 = \mathbf{I}$, $\mathbf{A}^6 = \mathbf{I}$, ...

So $\mathbf{A}^{2024} = \mathbf{I}$, so every point maps to itself.

5 a)

Rotation, anticlockwise about the origin, by 90°.

5 b)

$\mathbf{A}^2$ is a rotation about the origin by 180°.

$\mathbf{A}^3$ is a rotation, anticlockwise about the origin, by 270°.

$\mathbf{A}^4 = \mathbf{I}$.

$\mathbf{A}^5$ is a rotation, anticlockwise about the origin, by 90°.

$\mathbf{A}^6$ is a rotation about the origin by 180°.

$\mathbf{A}^7$ is a rotation, anticlockwise about the origin, by 270°.

$\mathbf{A}^8 = \mathbf{I}$.

...

$335 \div 4 = 83$ r 3.

So $\mathbf{A}^{335}$ is a rotation, anticlockwise about the origin, by 270°.

6 a)

A stretch, parallel to the x-axis, scale factor 2

Followed by

A stretch, parallel to the y-axis, scale factor 5

(either order)

$$\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$$

6 b)

An enlargement, centred at the origin, scale factor 3

Followed by

A reflection in the line $y = x$

(either order)

$$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ 3 & 0 \end{pmatrix}$$

6 c)

An enlargement, centred at the origin, scale factor 2

Followed by

A rotation, anticlockwise about the origin, by 90°

(either order)

$$\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$$

6 d)

An enlargement, centred at the origin, scale factor 5

Followed by

A reflection in the x-axis

(either order)

$$\begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & -5 \end{pmatrix}$$

6 e)

An enlargement, centred at the origin, scale factor 4

Followed by

A rotation, about the origin, by 180°

(either order)

$$\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -4 & 0 \\ 0 & -4 \end{pmatrix}$$

6 f)

An enlargement, centred at the origin, scale factor 8

Followed by

A reflection in the y-axis

(either order)

$$\begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -8 & 0 \\ 0 & 8 \end{pmatrix}$$

6 g)

A stretch, parallel to the y-axis, scale factor 9

Followed by

A rotation, anticlockwise about the origin, by 270° (or a rotation, clockwise about the origin, by 90°)

(either order)

$$\begin{pmatrix} 1 & 0 \\ 0 & 9 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -9 & 0 \end{pmatrix}$$

7)

$\det \mathbf{A} = 1 \times 2 - k \times 3 = 2 - 3k$

$\det \mathbf{B} = 5$

If both area and orientation are preserved, then $\det \mathbf{AB} = 1$.

$\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} \Rightarrow 1 = (2 - 3k) \times 5 \Rightarrow k = \frac{3}{5}$.

8)

$\det \mathbf{A} = (2k+1) \times (-2) - 4 \times 3k = -4k - 2 - 12k = -16k - 2$

$\det \mathbf{B} = -3$

If area is preserved but not orientation, then $\det \mathbf{AB} = -1$.

$\det \mathbf{AB} = \det \mathbf{A} \times \det \mathbf{B} \Rightarrow -1 = (-16k - 2) \times (-3) \Rightarrow k = -\frac{7}{48}$.

9)

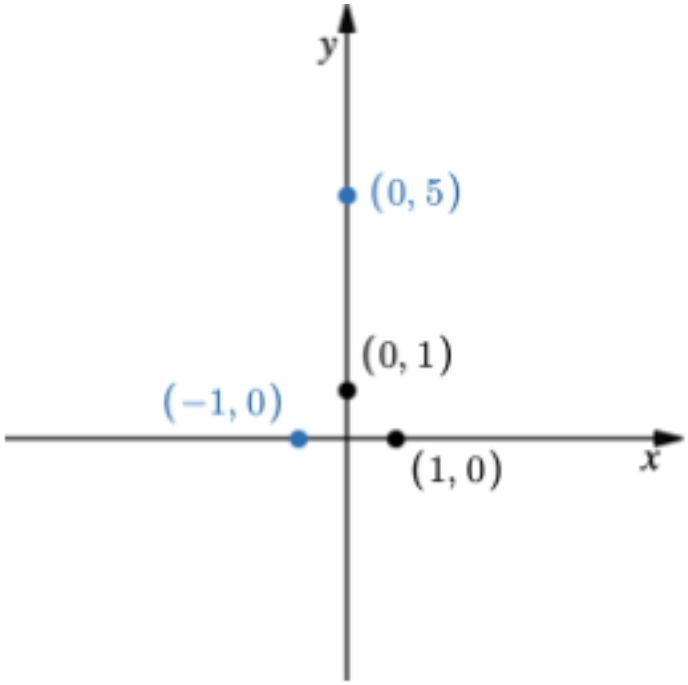
$$\mathbf{A}^2 = \begin{pmatrix} 100 & 0 \\ 0 & 100 \end{pmatrix} \Rightarrow \begin{pmatrix} 4 & p \\ q & p \end{pmatrix}^2 = \begin{pmatrix} 100 & 0 \\ 0 & 100 \end{pmatrix} \Rightarrow \begin{pmatrix} pq + 16 & p^2 + 4p \\ pq + 4q & p^2 + pq \end{pmatrix} = \begin{pmatrix} 100 & 0 \\ 0 & 100 \end{pmatrix}$$

$p^2 + 4p = 0 \Rightarrow p(p+4) = 0 \Rightarrow p = 0$ or $p = -4$.

$p \neq 0$ otherwise $pq + 16 \neq 100$.

So $p = -4$: $-4q + 16 = 100 \Rightarrow q = -21$.

10 a)



10 b)

A rotation, anticlockwise about the origin, by 90°

Followed by

A stretch, parallel to the y-axis, scale factor 5

$$\begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 5 & 0 \end{pmatrix}$$

11 a)

$$\begin{pmatrix} 1 & 0 \\ 0 & 10 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 40 \end{pmatrix}$$

11 b)

$$\begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

11 c)

$$\begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -3 \\ 1 & 0 \end{pmatrix}$$

11 d)

$$\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

11 e)

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$$

11 f)

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

11 g)

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

12)

$$\mathbf{D} = \mathbf{ABC} = \begin{pmatrix} 5 & -2 \\ 5 & -1 \end{pmatrix} \begin{pmatrix} \frac{4}{15} & -\frac{1}{3} \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -\frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{2}{3} \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$

So D is a reflection in the line $y = -x$.

13)

$$\mathbf{D} = \mathbf{CAB} = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} -4 & 7 \\ 1 & -2 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix} \begin{pmatrix} -4 & 7 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

So D is a rotation by $180°$ about the origin.

14)

$$\begin{pmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \cos 225° & -\sin 225° \\ \sin 225° & \cos 225° \end{pmatrix} \begin{pmatrix} \sqrt{2} & -3 & 0 \\ 2 & \sqrt{2} & -\sqrt{2} \end{pmatrix}$$

$$= \begin{pmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} \sqrt{2} & -3 & 0 \\ 2 & \sqrt{2} & -\sqrt{2} \end{pmatrix}$$

$$= \begin{pmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} -1+\sqrt{2} & 1+\frac{3}{2}\sqrt{2} & -1 \\ -1 & -1-\sqrt{2} & -1+\frac{3}{2}\sqrt{2} \end{pmatrix}$$

$$= \begin{pmatrix} \sqrt{2} & 0 \\ 0 & \sqrt{2} \end{pmatrix} \begin{pmatrix} 1+\sqrt{2} & 1-\frac{3}{2}\sqrt{2} & -1 \\ 1-\sqrt{2} & -1-\frac{3}{2}\sqrt{2} & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2+\sqrt{2} & -3+\sqrt{2} & -\sqrt{2} \\ -2+\sqrt{2} & -3-\sqrt{2} & \sqrt{2} \end{pmatrix}$$

So the coordinates of the vertices of triangle X' are $(2+\sqrt{2}, -2+\sqrt{2})$, $(-3+\sqrt{2}, -3-\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$.

15)

$$\begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 2\sqrt{2} \\ \sqrt{2} \end{pmatrix} = \begin{pmatrix} -8 \\ -24 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} k\cos\theta & -k\sin\theta \\ k\sin\theta & k\cos\theta \end{pmatrix} \begin{pmatrix} 2\sqrt{2} \\ \sqrt{2} \end{pmatrix} = \begin{pmatrix} -8 \\ -24 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2\sqrt{2}k\cos\theta - \sqrt{2}k\sin\theta \\ 2\sqrt{2}k\sin\theta + \sqrt{2}k\cos\theta \end{pmatrix} = \begin{pmatrix} -8 \\ -24 \end{pmatrix}$$

$2\sqrt{2}k\cos\theta - \sqrt{2}k\sin\theta = -8 \quad (1)$

$2\sqrt{2}k\sin\theta + \sqrt{2}k\cos\theta = -24 \quad (2)$

$2 \times (1) + (2) : 5\sqrt{2}k\cos\theta = -40 \Rightarrow \cos\theta = -\frac{40}{5\sqrt{2}k} = -\frac{4\sqrt{2}}{k}$

$(1) - 2 \times (2) : -5\sqrt{2}k\sin\theta = 40 \Rightarrow \sin\theta = -\frac{40}{5\sqrt{2}k} = -\frac{4\sqrt{2}}{k}$

$$k = \sqrt{(4\sqrt{2})^2 + (4\sqrt{2})^2} = 8$$

$\theta = \arctan 1 = 45°$.

As both $\cos\theta$ and $\sin\theta$ are negative, $\theta = 225°$.

So $k = 8$ and $\theta = 225°$.

16)

$$\begin{pmatrix} k & 0 \\ 0 & \frac{1}{k} \end{pmatrix} \begin{pmatrix} 1 & 4 \\ 2 & 6 \end{pmatrix} = \begin{pmatrix} k & 4k \\ \frac{2}{k} & \frac{6}{k} \end{pmatrix}$$

So $A'(k, \frac{2}{k})$ and $B'(4k, \frac{6}{k})$.

$$145 = (4k - k)^2 + (\frac{6}{k} - \frac{2}{k})^2$$

$$\Rightarrow 145 = (3k)^2 + (\frac{4}{k})^2$$

$$\Rightarrow 145 = 9k^2 + \frac{16}{k^2} \Rightarrow 145k^2 = 9k^4 + 16$$

$$\Rightarrow 9k^4 - 145k^2 + 16 = 0 \Rightarrow (9k^2 - 1)(k^2 - 16) = 0$$

$9k^2 - 1 = 0 \Rightarrow k^2 = \frac{1}{9} \Rightarrow k = \pm\frac{1}{3}$

$k^2 - 16 = 0 \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$

As k is a positive integer, we must have $k = 4$.

17)

$$\mathbf{C} = 3\begin{pmatrix} 2 & -5 \\ -7 & 4 \end{pmatrix} - \begin{pmatrix} 6 & -25 \\ -31 & 12 \end{pmatrix} = \begin{pmatrix} 6 & -15 \\ -21 & 12 \end{pmatrix} - \begin{pmatrix} 6 & -25 \\ -31 & 12 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 10 \\ 10 & 0 \end{pmatrix} = \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

This is a reflection in the line $y = x$, and an enlargement, centred at the origin, by scale factor 10.

18)

$$\mathbf{C} = \begin{pmatrix} 2 & 5 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} -100 & -40 \\ 40 & 20 \end{pmatrix} = \begin{pmatrix} 0 & 20 \\ -20 & 0 \end{pmatrix} = \begin{pmatrix} 20 & 0 \\ 0 & 20 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

This is a rotation, anticlockwise about the origin, by 270°, and an enlargement, centred at the origin, by scale factor 10.

19)

$$\mathbf{A}^2 = \begin{pmatrix} 2k & m \\ k & 4 \end{pmatrix} \begin{pmatrix} 2k & m \\ k & 4 \end{pmatrix} = \begin{pmatrix} 4k^2 + km & 2km + 4m \\ 2k^2 + 4k & km + 16 \end{pmatrix}$$

$4k^2 + km = 11 \quad (1)$

$km + 16 = 11 \quad (2)$

$\therefore km = -5$

$4k^2 - 5 = 11 \Rightarrow k^2 = 4 \Rightarrow k = \pm 2$

If $k = 2$ then $m = -\frac{5}{2}$.

If $k = -2$ then $m = \frac{5}{2}$.

$2k^2 + 4k = 0 \quad (3)$

Check in (3):

$k = 2$, $m = -\frac{5}{2}$: $2 \times 2^2 + 4 \times 2 = 16 \neq 0$.

$k = -2$, $m = \frac{5}{2}$: $2 \times (-2)^2 + 4 \times (-2) = 0$.

$2km + 4m = 0 \quad (4)$

Check in (4):

$k = -2$, $m = \frac{5}{2}$: $2 \times (-2) \times \frac{5}{2} + 4 \times \frac{5}{2} = 0$.

So $k = -2$ and $m = \frac{5}{2}$.

20)

$$\begin{pmatrix} -5 & 4k \\ -2k & p \end{pmatrix}^2 = \begin{pmatrix} 9 & 0 \\ 0 & 9 \end{pmatrix} \Rightarrow \begin{pmatrix} 25 - 8k^2 & 4kp - 20k \\ -2kp + 10k & p^2 - 8k^2 \end{pmatrix} = \begin{pmatrix} 9 & 0 \\ 0 & 9 \end{pmatrix}$$

$25 - 8k^2 = 9 \Rightarrow 8k^2 = 16 \Rightarrow k^2 = 2 \Rightarrow k = \pm\sqrt{2}$.

As k is a positive constant, we must have $k = \sqrt{2}$.

$p^2 - 8k^2 = 9 \Rightarrow p^2 - 8(\sqrt{2})^2 = 9 \Rightarrow p^2 = 25 \Rightarrow p = \pm 5$.

As p is a positive constant, we must have $p = 5$.

So $k = \sqrt{2}$ and $p = 5$.

21)

$$\begin{pmatrix} 3 & -5 \\ 5 & 1 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 12 & 0 \\ 0 & 12 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 3a - 5c & 3b - 5d \\ 5a + c & 5b + d \end{pmatrix} = \begin{pmatrix} 0 & 12 \\ -12 & 0 \end{pmatrix}$$

$3a - 5c = 0 \quad (1)$

$5a + c = -12 \quad (2)$

$\therefore a = -\frac{15}{7}$ and $c = -\frac{9}{7}$.

$3b - 5d = 12 \quad (3)$

$5b + d = 0 \quad (4)$

$\therefore b = \frac{3}{7}$ and $d = -\frac{15}{7}$.

So $a = -\frac{15}{7}$, $b = \frac{3}{7}$, $c = -\frac{9}{7}$ and $d = -\frac{15}{7}$.

22 a)

$$\mathbf{A}^3 = \begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix}^3 = \begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix} \begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix} \begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix}$$

$$= \begin{pmatrix} -3\sqrt{3} & -9 \\ 9 & -3\sqrt{3} \end{pmatrix} \begin{pmatrix} -54 & 54\sqrt{3} \\ -54\sqrt{3} & -54 \end{pmatrix} = \begin{pmatrix} 648\sqrt{3} & 0 \\ 0 & 648\sqrt{3} \end{pmatrix}$$

$\therefore \mathbf{A}^3 = 648\sqrt{3}\mathbf{I}$.

22 b)

It is an enlargement, centred at the origin, by scale factor $648\sqrt{3}$.

22 c)

It is an enlargement, centred at the origin, by scale factor $\sqrt[3]{648\sqrt{3}} = 6\sqrt{3}$, and a rotation by $\frac{360}{3} = 120°$, anticlockwise about the origin.

23 a)

$$\mathbf{A}^4 = \begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}^4 = \begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}\begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}\begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}\begin{pmatrix} 0 & 3\sqrt{2} \\ -3\sqrt{2} & 0 \end{pmatrix}$$
$$= \begin{pmatrix} -18 & 0 \\ 0 & -18 \end{pmatrix}\begin{pmatrix} -18 & 0 \\ 0 & -18 \end{pmatrix} = \begin{pmatrix} 324 & 0 \\ 0 & 324 \end{pmatrix}$$

$\therefore \mathbf{A}^4 = 324\mathbf{I}$.

23 b)

It is an enlargement, centred at the origin, by scale factor 324.

23 c)

It is an enlargement, centred at the origin, by scale factor $\sqrt[4]{324} = 3\sqrt{2}$, and a rotation by $\frac{360}{4} = 90°$, anticlockwise about the origin.

23 d)

It is an enlargement, centred at the origin, by scale factor $\frac{1}{3\sqrt{2}}$, and a rotation by 90°, clockwise about the origin.

24 a)

$$\mathbf{A}^2 = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}^2 = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}\begin{pmatrix} a & -b \\ b & a \end{pmatrix} = \begin{pmatrix} a^2 - b^2 & -2ab \\ 2ab & a^2 - b^2 \end{pmatrix}$$

24 b)

$$a^2 - b^2 = \left(\frac{1+\sqrt{3}}{4}\right)^2 - \left(\frac{\sqrt{3}-1}{4}\right)^2 = \frac{4+2\sqrt{3}}{16} - \frac{4-2\sqrt{3}}{16} = \frac{4\sqrt{3}}{16} = \frac{\sqrt{3}}{4}$$
$$2ab = 2\left(\frac{1+\sqrt{3}}{4}\right)\left(\frac{\sqrt{3}-1}{4}\right) = 2\left(\frac{2}{16}\right) = \frac{1}{4}$$
$$\mathbf{A}^2 = \begin{pmatrix} \frac{\sqrt{3}}{4} & -\frac{1}{4} \\ \frac{1}{4} & \frac{\sqrt{3}}{4} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$$

$\therefore k = \frac{1}{2}$.

24 c)

i)

$$\mathbf{A}^6 = \begin{pmatrix} \frac{1+\sqrt{3}}{4} & -\frac{\sqrt{3}-1}{4} \\ \frac{\sqrt{3}-1}{4} & \frac{1+\sqrt{3}}{4} \end{pmatrix}^6 = \begin{pmatrix} 0 & -\frac{1}{8} \\ \frac{1}{8} & 0 \end{pmatrix} = \frac{1}{8}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

This is a rotation, anticlockwise about the origin, by 90°, and an enlargement, centred at the origin, by scale factor $\frac{1}{8}$.

ii)

$$\mathbf{A}^{12} = \begin{pmatrix} \frac{1+\sqrt{3}}{4} & -\frac{\sqrt{3}-1}{4} \\ \frac{\sqrt{3}-1}{4} & \frac{1+\sqrt{3}}{4} \end{pmatrix}^{12} = \begin{pmatrix} -\frac{1}{64} & 0 \\ 0 & -\frac{1}{64} \end{pmatrix} = \frac{1}{64}\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

This is a rotation, about the origin, by 180°, and an enlargement, centred at the origin, by scale factor $\frac{1}{64}$.

iii)

$$\mathbf{A}^{24} = \begin{pmatrix} \frac{1+\sqrt{3}}{4} & -\frac{\sqrt{3}-1}{4} \\ \frac{\sqrt{3}-1}{4} & \frac{1+\sqrt{3}}{4} \end{pmatrix}^{24} = \begin{pmatrix} \frac{1}{4096} & 0 \\ 0 & \frac{1}{4096} \end{pmatrix} = \frac{1}{4096}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

This is an enlargement, centred at the origin, by scale factor $\frac{1}{4096}$.

iv)

$$\mathbf{A} = \begin{pmatrix} \frac{1+\sqrt{3}}{4} & -\frac{\sqrt{3}-1}{4} \\ \frac{\sqrt{3}-1}{4} & \frac{1+\sqrt{3}}{4} \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} \frac{\sqrt{2}+\sqrt{6}}{4} & -\frac{\sqrt{6}-\sqrt{2}}{4} \\ \frac{\sqrt{6}-\sqrt{2}}{4} & \frac{\sqrt{2}+\sqrt{6}}{4} \end{pmatrix}$$

This is a rotation, about the origin, by 15°, and an enlargement, centred at the origin, by scale factor $\frac{1}{\sqrt{2}}$.

25 a)

T: Reflection in the line $y = \frac{\sqrt{3}}{3}x$:

$\theta = \arctan\frac{\sqrt{3}}{3} = 30°$.

Matrix is

$$\begin{pmatrix} \cos(2\times 30°) & \sin(2\times 30°) \\ \sin(2\times 30°) & -\cos(2\times 30°) \end{pmatrix} = \begin{pmatrix} \cos 60° & \sin 60° \\ \sin 60° & -\cos 60° \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

U: Reflection in the line $y = \sqrt{3}x$:

$\theta = \arctan\sqrt{3} = 60°$.

Matrix is

$$\begin{pmatrix} \cos(2\times 60°) & \sin(2\times 60°) \\ \sin(2\times 60°) & -\cos(2\times 60°) \end{pmatrix} = \begin{pmatrix} \cos 120° & \sin 120° \\ \sin 120° & -\cos 120° \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

T followed by U:

$$\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

25 b)

U followed by T:

$$\begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}\begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

26 a)

$\cos\theta = 0.28 \Rightarrow \theta = \{73.7°, 286°\}$ (3 SF)

$\sin\theta = 0.96 \Rightarrow \theta = \{73.7°, 106°\}$ (3 SF)

A represents a rotation, anticlockwise about the origin, by 73.7° (3 SF).

26 b)

$\cos 2\theta = -0.8432 \Rightarrow \theta = \{73.7°, 106°\}$ (3 SF)

$\sin 2\theta = 0.5376 \Rightarrow \theta = \{16.3°, 73.7°\}$ (3 SF)

B represents a reflection in the line $y = (\tan 73.7°)x \Rightarrow y = 3.43x$ (3 SF).

26 c)

$$\mathbf{C} = \begin{pmatrix} 0.28 & -0.96 \\ 0.96 & 0.28 \end{pmatrix} \begin{pmatrix} -0.8432 & 0.5376 \\ 0.5376 & 0.8432 \end{pmatrix} = \begin{pmatrix} -0.752192 & -0.658944 \\ -0.658944 & 0.752192 \end{pmatrix}$$

$\cos 2\theta = -0.752192 \Rightarrow \theta = \{69.4°, 111°\}$ (3 SF)

$\sin 2\theta = -0.658944 \Rightarrow \theta = \{111°, 159°\}$ (3 SF)

$\mathbf{C}$ represents a reflection in the line $y = (\tan 111°)x \Rightarrow y = -2.61x$ (3 SF).

27)

$y\sqrt{3} - 3x = 0 \Rightarrow y = \frac{3}{\sqrt{3}}x \Rightarrow y = \sqrt{3}x.$

$\tan\theta = \sqrt{3} \Rightarrow \theta = \arctan\sqrt{3} = 60°.$

$$\therefore \begin{pmatrix} \cos(2\times 60°) & \sin(2\times 60°) \\ \sin(2\times 60°) & -\cos(2\times 60°) \end{pmatrix} = \begin{pmatrix} \cos 120° & \sin 120° \\ \sin 120° & -\cos 120° \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$$

$y\sqrt{3} + x = 0 \Rightarrow y = -\frac{1}{\sqrt{3}}x \Rightarrow y = -\frac{\sqrt{3}}{3}x.$

$\tan\theta = -\frac{\sqrt{3}}{3} \Rightarrow \theta = 180° + \arctan(-\frac{\sqrt{3}}{3}) = 150°.$

$$\therefore \begin{pmatrix} \cos(2\times 150°) & \sin(2\times 150°) \\ \sin(2\times 150°) & -\cos(2\times 150°) \end{pmatrix} = \begin{pmatrix} \cos 300° & \sin 300° \\ \sin 300° & -\cos 300° \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

This is a reflection in the line $y = -x$.

28 a)

$$\mathbf{A}^2 = \begin{pmatrix} 3 & 3 \\ 3 & -3 \end{pmatrix}^2 = \begin{pmatrix} 18 & 0 \\ 0 & 18 \end{pmatrix}$$

It is an enlargement, centred at the origin, by scale factor 18.

28 b)

It is an enlargement, centred at the origin, by scale factor $\sqrt{18} = 3\sqrt{2}$.

$$\frac{1}{3\sqrt{2}}\mathbf{A} = \frac{1}{3\sqrt{2}} \begin{pmatrix} 3 & 3 \\ 3 & -3 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \end{pmatrix}$$

$\cos 2\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{22.5°, 157.5°\}$

$\sin 2\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{22.5°, 67.5°\}$

So it is a reflection in the line $y = (\tan 22.5°)x \Rightarrow y = 0.414x$ (3 SF).

29)

$$\begin{pmatrix} -\cos 2\theta & -\sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -\sqrt{2} \end{pmatrix} \Rightarrow \begin{pmatrix} -\cos 2\theta + \sin 2\theta \\ -\sin 2\theta - \cos 2\theta \end{pmatrix} = \begin{pmatrix} 0 \\ -\sqrt{2} \end{pmatrix}$$

$-\cos 2\theta + \sin 2\theta = 0 \Rightarrow \sin 2\theta = \cos 2\theta \Rightarrow \frac{\sin 2\theta}{\cos 2\theta} = 1 \Rightarrow \tan 2\theta = 1.$

$\Rightarrow \theta = \{22.5°, 112.5°, 202.5°, 292.5°\}.$

If $\theta = 22.5°$, then $-\sin(2 \times 22.5°) - \cos(2 \times 22.5°) = -\sin 45° - \cos 45° = -\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = -\sqrt{2}.$

So $\theta = 22.5°$ is the smallest value of θ.

30)

$$\begin{pmatrix} 3\cos\theta & -\sin\theta \\ \sin\theta & 3\cos\theta \end{pmatrix} \begin{pmatrix} 3 \\ -\frac{1}{\sqrt{3}} \end{pmatrix} = \begin{pmatrix} k \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 9\cos\theta + \frac{1}{\sqrt{3}}\sin\theta \\ 3\sin\theta - \frac{3}{\sqrt{3}}\cos\theta \end{pmatrix} = \begin{pmatrix} k \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 9\cos\theta + \frac{\sqrt{3}}{3}\sin\theta \\ 3\sin\theta - \sqrt{3}\cos\theta \end{pmatrix} = \begin{pmatrix} k \\ 0 \end{pmatrix}$$

$3\sin\theta - \sqrt{3}\cos\theta = 0 \Rightarrow 3\sin\theta = \sqrt{3}\cos\theta \Rightarrow \frac{3\sin\theta}{\cos\theta} = \sqrt{3} \Rightarrow 3\tan\theta = \sqrt{3}.$

$\Rightarrow \tan\theta = \frac{\sqrt{3}}{3} \Rightarrow \theta = 30°$ as θ is a positive acute angle.

$k = 9\cos 30° + \frac{\sqrt{3}}{3}\sin 30° = \frac{9\sqrt{3}}{2} + \frac{\sqrt{3}}{6} = \frac{14\sqrt{3}}{3}.$

31 a)

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

So it is a rotation by 40°, anticlockwise about the origin.

31 b)

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

$\cos\theta = \cos 40° \Rightarrow \theta = \{40°, 320°\}$

$\sin\theta = -\sin 40° \Rightarrow \theta = \{220°, 320°\}$

So it is a rotation by 320°, anticlockwise about the origin, or a rotation by 40°, clockwise about the origin.

31 c)

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

So it is a reflection in the line $y = (\tan 20°)x \Rightarrow y = 0.364x$ (3 SF).

31 d)

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

$\cos 2\theta = -\cos 40° \Rightarrow \theta = \{70°, 110°\}$

$\sin 2\theta = \sin 40° \Rightarrow \theta = \{20°, 70°\}$

So it is a reflection in the line $y = (\tan 70°)x \Rightarrow y = 2.75x$ (3 SF).

32 a)

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

$\cos\theta = -\cos 40^\circ \Rightarrow \theta = \{140^\circ, 220^\circ\}$

$\sin\theta = \sin 40^\circ \Rightarrow \theta = \{40^\circ, 140^\circ\}$

So it is a rotation by 140°, anticlockwise about the origin.

32 b)

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

$\cos\theta = -\cos 40^\circ \Rightarrow \theta = \{140^\circ, 220^\circ\}$

$\sin\theta = -\sin 40^\circ \Rightarrow \theta = \{220^\circ, 320^\circ\}$

So it is a rotation by 220°, anticlockwise about the origin.

32 c)

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

$\cos 2\theta = \cos 40^\circ \Rightarrow \theta = \{20^\circ, 160^\circ\}$

$\sin 2\theta = -\sin 40^\circ \Rightarrow \theta = \{110^\circ, 160^\circ\}$

So it is a reflection in the line $y = (\tan 160^\circ)x \Rightarrow y = -0.364x$ (3 SF).

32 d)

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

$\cos 2\theta = -\cos 40^\circ \Rightarrow \theta = \{70^\circ, 110^\circ\}$

$\sin 2\theta = -\sin 40^\circ \Rightarrow \theta = \{110^\circ, 160^\circ\}$

So it is a reflection in the line $y = (\tan 110^\circ)x \Rightarrow y = -2.75x$ (3 SF).

33 a)

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

So it is a rotation by 140°, anticlockwise about the origin.

33 b)

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

$\cos\theta = \cos 140^\circ \Rightarrow \theta = \{140^\circ, 220^\circ\}$

$\sin\theta = -\sin 140^\circ \Rightarrow \theta = \{220^\circ, 320^\circ\}$

So it is a rotation by 220°, anticlockwise about the origin.

33 c)

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

So it is a reflection in the line $y = (\tan 70°)x \Rightarrow y = 2.75x$ (3 SF).

33 d)

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

$\cos 2\theta = -\cos 140° \Rightarrow \theta = \{20°, 160°\}$

$\sin 2\theta = \sin 140° \Rightarrow \theta = \{20°, 70°\}$

So it is a reflection in the line $y = (\tan 20°)x \Rightarrow y = 0.364x$ (3 SF).

34 a)

$\sin 40° \equiv \cos 50°$ and $\cos 40° \equiv \sin 50°$.

So $\begin{pmatrix} \sin 40° & -\cos 40° \\ \cos 40° & \sin 40° \end{pmatrix} = \begin{pmatrix} \cos 50° & -\sin 50° \\ \sin 50° & \cos 50° \end{pmatrix}$.

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

So it is a rotation by 50°, anticlockwise about the origin.

34 b)

$\sin 40° \equiv \cos 50°$ and $\cos 40° \equiv \sin 50°$.

So $\begin{pmatrix} \sin 40° & \cos 40° \\ -\cos 40° & \sin 40° \end{pmatrix} = \begin{pmatrix} \cos 50° & \sin 50° \\ -\sin 50° & \cos 50° \end{pmatrix}$.

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

$\cos\theta = \cos 50° \Rightarrow \theta = \{50°, 310°\}$

$\sin\theta = -\sin 50° \Rightarrow \theta = \{230°, 310°\}$

So it is a rotation by 310°, anticlockwise about the origin, or a rotation by 50°, clockwise about the origin.

34 c)

$\sin 40° \equiv \cos 50°$ and $\cos 40° \equiv \sin 50°$.

So $\begin{pmatrix} \sin 40° & \cos 40° \\ \cos 40° & -\sin 40° \end{pmatrix} = \begin{pmatrix} \cos 50° & \sin 50° \\ \sin 50° & -\cos 50° \end{pmatrix}$.

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

So it is a reflection in the line $y = (\tan 25°)x \Rightarrow y = 0.466x$ (3 SF).

34 d)

$\sin 40° \equiv \cos 50°$ and $\cos 40° \equiv \sin 50°$.

So $\begin{pmatrix} -\sin 40° & \cos 40° \\ \cos 40° & \sin 40° \end{pmatrix} = \begin{pmatrix} -\cos 50° & \sin 50° \\ \sin 50° & \cos 50° \end{pmatrix}$.

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

$\cos 2\theta = -\cos 50^\circ \Rightarrow \theta = \{65^\circ, 115^\circ\}$

$\sin 2\theta = \sin 50^\circ \Rightarrow \theta = \{25^\circ, 65^\circ\}$

So it is a reflection in the line $y = (\tan 65^\circ)x \Rightarrow y = 2.14x$ (3 SF).

35 a)

$\sin 140^\circ \equiv \cos 310^\circ$ and $\cos 140^\circ \equiv \sin 310^\circ$.

So $\begin{pmatrix} \sin 140^\circ & -\cos 140^\circ \\ \cos 140^\circ & \sin 140^\circ \end{pmatrix} = \begin{pmatrix} \cos 310^\circ & -\sin 310^\circ \\ \sin 310^\circ & \cos 310^\circ \end{pmatrix}$.

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

So it is a rotation by 310°, anticlockwise about the origin, or a rotation by 50°, clockwise about the origin.

Alternative Method:

$\sin 140^\circ \equiv \cos 50^\circ$ and $\cos 140^\circ \equiv -\sin 50^\circ$.

So $\begin{pmatrix} \sin 140^\circ & -\cos 140^\circ \\ \cos 140^\circ & \sin 140^\circ \end{pmatrix} = \begin{pmatrix} \cos 50^\circ & \sin 50^\circ \\ -\sin 50^\circ & \cos 50^\circ \end{pmatrix}$.

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

$\cos\theta = \cos 50^\circ \Rightarrow \theta = \{50^\circ, 310^\circ\}$

$\sin\theta = -\sin 50^\circ \Rightarrow \theta = \{230^\circ, 310^\circ\}$

So it is a rotation by 310°, anticlockwise about the origin, or a rotation by 50°, clockwise about the origin.

35 b)

$\sin 140^\circ \equiv \cos 310^\circ$ and $\cos 140^\circ \equiv \sin 310^\circ$.

So $\begin{pmatrix} \sin 140^\circ & \cos 140^\circ \\ -\cos 140^\circ & \sin 140^\circ \end{pmatrix} = \begin{pmatrix} \cos 310^\circ & \sin 310^\circ \\ -\sin 310^\circ & \cos 310^\circ \end{pmatrix}$.

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

$\cos\theta = \cos 310^\circ \Rightarrow \theta = \{50^\circ, 310^\circ\}$

$\sin\theta = -\sin 310^\circ \Rightarrow \theta = \{50^\circ, 130^\circ\}$

So it is a rotation by 50°, anticlockwise about the origin.

Alternative Method:

$\sin 140^\circ \equiv \cos 50^\circ$ and $\cos 140^\circ \equiv -\sin 50^\circ$.

So $\begin{pmatrix} \sin 140^\circ & \cos 140^\circ \\ -\cos 140^\circ & \sin 140^\circ \end{pmatrix} = \begin{pmatrix} \cos 50^\circ & -\sin 50^\circ \\ \sin 50^\circ & \cos 50^\circ \end{pmatrix}$.

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

So it is a rotation by 50°, anticlockwise about the origin, or a rotation by 310°, anticlockwise about the origin.

35 c)

$\sin 140° \equiv \cos 310°$ and $\cos 140° \equiv \sin 310°$.

So $\begin{pmatrix} \sin 140° & \cos 140° \\ \cos 140° & -\sin 140° \end{pmatrix} = \begin{pmatrix} \cos 310° & \sin 310° \\ \sin 310° & -\cos 310° \end{pmatrix}$.

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

So it is a reflection in the line $y = (\tan 155°)x$, which is roughly $y = -0.466x$.

Alternative Method:

$\sin 140° \equiv \cos 50°$ and $\cos 140° \equiv -\sin 50°$.

So $\begin{pmatrix} \sin 140° & \cos 140° \\ \cos 140° & -\sin 140° \end{pmatrix} = \begin{pmatrix} \cos 50° & -\sin 50° \\ -\sin 50° & -\cos 50° \end{pmatrix}$.

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

$\cos 2\theta = \cos 50° \Rightarrow \theta = \{25°, 155°, 205°, 335°\}$

$\sin 2\theta = -\sin 50° \Rightarrow \theta = \{115°, 155°, 295°, 335°\}$

So it is a reflection in the line $y = (\tan 155°)x$, which is roughly $y = -0.466x$.

35 d)

$\sin 140° \equiv \cos 310°$ and $\cos 140° \equiv \sin 310°$.

So $\begin{pmatrix} -\sin 140° & \cos 140° \\ \cos 140° & \sin 140° \end{pmatrix} = \begin{pmatrix} -\cos 310° & \sin 310° \\ \sin 310° & \cos 310° \end{pmatrix}$.

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

$\cos 2\theta = -\cos 310° \Rightarrow \theta = \{65°, 115°\}$

$\sin 2\theta = \sin 310° \Rightarrow \theta = \{115°, 155°\}$

So it is a reflection in the line $y = (\tan 115°)x$, which is roughly $y = -2.14x$.

Alternative Method:

$\sin 140° \equiv \cos 50°$ and $\cos 140° \equiv -\sin 50°$.

So $\begin{pmatrix} -\sin 140° & \cos 140° \\ \cos 140° & \sin 140° \end{pmatrix} = \begin{pmatrix} -\cos 50° & -\sin 50° \\ -\sin 50° & \cos 50° \end{pmatrix}$.

This is similar to the general reflection matrix $\begin{pmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{pmatrix}$.

$\cos 2\theta = -\cos 50° \Rightarrow \theta = \{65°, 115°, 245°, 295°\}$

$\sin 2\theta = -\sin 50° \Rightarrow \theta = \{115°, 155°, 295°, 335°\}$

So it is a reflection in the line $y = (\tan 115°)x$, which is roughly $y = -0.466x$.

36)

$$\begin{pmatrix} p & -q \\ q & p \end{pmatrix}^2 = \begin{pmatrix} \cos 150° & -\sin 150° \\ \sin 150° & \cos 150° \end{pmatrix} \Rightarrow \begin{pmatrix} p^2 - q^2 & -2pq \\ 2pq & p^2 - q^2 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$$

$2pq = \frac{1}{2} \Rightarrow q = \frac{1}{4p}$

$p^2 - q^2 = -\frac{\sqrt{3}}{2} \Rightarrow p^2 - (\frac{1}{4p})^2 = -\frac{\sqrt{3}}{2} \Rightarrow p^2 - \frac{1}{16p^2} = -\frac{\sqrt{3}}{2}$

$\Rightarrow 16p^4 - 1 = -8\sqrt{3}p^2$

$\Rightarrow 16p^4 + 8\sqrt{3}p^2 - 1 = 0$

$$\Rightarrow p^2 = \frac{-8\sqrt{3} \pm \sqrt{(8\sqrt{3})^2 - 4 \times 16 \times (-1)}}{32}$$

$$\Rightarrow p^2 = \frac{-8\sqrt{3} \pm 16}{32} \Rightarrow p^2 = \frac{-\sqrt{3} \pm 2}{4}$$

$-\sqrt{3} - 2 < 0$ so $p^2 = \frac{-\sqrt{3}+2}{4} \Rightarrow p = \sqrt{\frac{2-\sqrt{3}}{4}}$.

$$\begin{pmatrix} \cos 75^\circ & -\sin 75^\circ \\ \sin 75^\circ & \cos 75^\circ \end{pmatrix}^2 = \begin{pmatrix} \cos 150^\circ & -\sin 150^\circ \\ \sin 150^\circ & \cos 150^\circ \end{pmatrix}$$

$\therefore \cos 75^\circ = \sqrt{\frac{2-\sqrt{3}}{4}}$.

37)

$$\begin{pmatrix} p & -q \\ q & p \end{pmatrix}^2 = \begin{pmatrix} \cos 105^\circ & -\sin 105^\circ \\ \sin 105^\circ & \cos 105^\circ \end{pmatrix} \Rightarrow \begin{pmatrix} p^2 - q^2 & -2pq \\ 2pq & p^2 - q^2 \end{pmatrix} = \begin{pmatrix} \frac{-\sqrt{6}+\sqrt{2}}{4} & -\frac{\sqrt{6}+\sqrt{2}}{4} \\ \frac{\sqrt{6}+\sqrt{2}}{4} & \frac{-\sqrt{6}+\sqrt{2}}{4} \end{pmatrix}$$

$2pq = \frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow p = \frac{\sqrt{6}+\sqrt{2}}{8q}$

$p^2 - q^2 = \frac{-\sqrt{6}+\sqrt{2}}{4} \Rightarrow (\frac{\sqrt{6}+\sqrt{2}}{8q})^2 - q^2 = \frac{-\sqrt{6}+\sqrt{2}}{4} \Rightarrow \frac{8+4\sqrt{3}}{64q^2} - q^2 = \frac{-\sqrt{6}+\sqrt{2}}{4}$

$\Rightarrow 8 + 4\sqrt{3} - 64q^4 = 16(-\sqrt{6} + \sqrt{2})q^2$

$\Rightarrow 64q^4 + 16(-\sqrt{6} + \sqrt{2})q^2 - 4\sqrt{3} - 8 = 0$

$\Rightarrow 16q^4 + 4(-\sqrt{6} + \sqrt{2})q^2 - \sqrt{3} - 2 = 0$

$$\Rightarrow q^2 = \frac{-4(-\sqrt{6} + \sqrt{2}) \pm \sqrt{(-4(-\sqrt{6} + \sqrt{2}))^2 - 4 \times 16 \times (-\sqrt{3} - 2)}}{32}$$

$$\Rightarrow q^2 = \frac{-4(-\sqrt{6} + \sqrt{2}) \pm 16}{32} \Rightarrow q^2 = \frac{\sqrt{6} - \sqrt{2} \pm 4}{8}$$

$\sqrt{6} - \sqrt{2} - 4 < 0$ so $q^2 = \frac{\sqrt{6}-\sqrt{2}+4}{8}$

$\Rightarrow q = \sqrt{\frac{\sqrt{6}-\sqrt{2}+4}{8}}$.

$$\begin{pmatrix} \cos 52.5^\circ & -\sin 52.5^\circ \\ \sin 52.5^\circ & \cos 52.5^\circ \end{pmatrix}^2 = \begin{pmatrix} \cos 105^\circ & -\sin 105^\circ \\ \sin 105^\circ & \cos 105^\circ \end{pmatrix}$$

$\therefore \sin 52.5^\circ = \sqrt{\frac{\sqrt{6}-\sqrt{2}+4}{8}}$.

38 a)

$\det \mathbf{A} = 10 \times 10 - (-10) \times 10 = 200$.

$\sqrt{200} = 10\sqrt{2}$.

$$\begin{pmatrix} 10\sqrt{2} & 0 \\ 0 & 10\sqrt{2} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 10 & -10 \\ 10 & 10 \end{pmatrix} \Rightarrow \begin{pmatrix} 10\sqrt{2}a & 10\sqrt{2}b \\ 10\sqrt{2}c & 10\sqrt{2}d \end{pmatrix} = \begin{pmatrix} 10 & -10 \\ 10 & 10 \end{pmatrix}$$

$10\sqrt{2}a = 10 \Rightarrow a = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

$10\sqrt{2}b = -10 \Rightarrow b = -\frac{1}{\sqrt{2}} = -\frac{\sqrt{2}}{2}$

$10\sqrt{2}c = 10 \Rightarrow c = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

$10\sqrt{2}d = 10 \Rightarrow d = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

So $\mathbf{A} = \begin{pmatrix} 10\sqrt{2} & 0 \\ 0 & 10\sqrt{2} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$.

$\cos\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 315°\}$

$\sin\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 135°\}$

So **A** represents an enlargement, centred at the origin, by scale factor $10\sqrt{2}$, and a rotation, anti-clockwise about the origin, by scale factor 45° (either order).

38 b)

$\det \mathbf{A} = -8\sqrt{3} \times 8\sqrt{3} - 0 \times 0 = -192.$

$\sqrt{192} = 8\sqrt{3}.$

$$\begin{pmatrix} 8\sqrt{3} & 0 \\ 0 & 8\sqrt{3} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} -8\sqrt{3} & 0 \\ 0 & 8\sqrt{3} \end{pmatrix} \Rightarrow \begin{pmatrix} 8\sqrt{3}a & 8\sqrt{3}b \\ 8\sqrt{3}c & 8\sqrt{3}d \end{pmatrix} = \begin{pmatrix} -8\sqrt{3} & 0 \\ 0 & 8\sqrt{3} \end{pmatrix}$$

$8\sqrt{3}a = -8\sqrt{3} \Rightarrow a = -1$

$8\sqrt{3}b = 0 \Rightarrow b = 0$

$8\sqrt{3}c = 0 \Rightarrow c = 0$

$8\sqrt{3}d = 8\sqrt{3} \Rightarrow d = 1$

So $\mathbf{A} = \begin{pmatrix} 8\sqrt{3} & 0 \\ 0 & 8\sqrt{3} \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.

So **A** represents an enlargement, centred at the origin, by scale factor $8\sqrt{3}$, and a reflection in the y-axis (either order).

38 c)

$\det \mathbf{A} = 10\sqrt{3} \times 10\sqrt{3} - (-10) \times 10 = 400.$

$\sqrt{400} = 20.$

$$\begin{pmatrix} 20 & 0 \\ 0 & 20 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 10\sqrt{3} & -10 \\ 10 & 10\sqrt{3} \end{pmatrix} \Rightarrow \begin{pmatrix} 20a & 20b \\ 20c & 20d \end{pmatrix} = \begin{pmatrix} 10\sqrt{3} & -10 \\ 10 & 10\sqrt{3} \end{pmatrix}$$

$20a = 10\sqrt{3} \Rightarrow a = \frac{\sqrt{3}}{2}$

$20b = -10 \Rightarrow b = -\frac{1}{2}$

$20c = 10 \Rightarrow c = \frac{1}{2}$

$20d = 10\sqrt{3} \Rightarrow d = \frac{\sqrt{3}}{2}$

So $\mathbf{A} = \begin{pmatrix} 20 & 0 \\ 0 & 20 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$.

$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{30°, 330°\}$

$\sin\theta = \frac{1}{2} \Rightarrow \theta = \{30°, 150°\}$

So **A** represents an enlargement, centred at the origin, by scale factor 20, and a rotation, anticlockwise about the origin, by scale factor 30° (either order).

38 d)

$\det \mathbf{A} = \sqrt{5} \times \sqrt{5} - \sqrt{5} \times (-\sqrt{5}) = 10.$

$$\begin{pmatrix} \sqrt{10} & 0 \\ 0 & \sqrt{10} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \sqrt{5} & \sqrt{5} \\ -\sqrt{5} & \sqrt{5} \end{pmatrix} \Rightarrow \begin{pmatrix} \sqrt{10}a & \sqrt{10}b \\ \sqrt{10}c & \sqrt{10}d \end{pmatrix} = \begin{pmatrix} \sqrt{5} & \sqrt{5} \\ -\sqrt{5} & \sqrt{5} \end{pmatrix}$$

$\sqrt{10}a = \sqrt{5} \Rightarrow a = \frac{\sqrt{2}}{2}$

$\sqrt{10}b = \sqrt{5} \Rightarrow b = \frac{\sqrt{2}}{2}$

$\sqrt{10}c = -\sqrt{5} \Rightarrow c = -\frac{\sqrt{2}}{2}$

$\sqrt{10}d = \sqrt{5} \Rightarrow d = \frac{\sqrt{2}}{2}$

So $\mathbf{A} = \begin{pmatrix} \sqrt{10} & 0 \\ 0 & \sqrt{10} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$.

$\cos\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 315°\}$

$\sin\theta = -\frac{\sqrt{2}}{2} \Rightarrow \theta = \{225°, 315°\}$

So $\mathbf{A}$ represents an enlargement, centred at the origin, by scale factor $\sqrt{10}$, and a rotation, anti-clockwise about the origin, by 315° (either order).

38 e)

$\det \mathbf{A} = 5\sqrt{30} \times 5\sqrt{30} - 5\sqrt{10} \times (-5\sqrt{10}) = 1000.$

$$\begin{pmatrix} 10\sqrt{10} & 0 \\ 0 & 10\sqrt{10} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 5\sqrt{30} & 5\sqrt{10} \\ -5\sqrt{10} & 5\sqrt{30} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 10\sqrt{10}a & 10\sqrt{10}b \\ 10\sqrt{10}c & 10\sqrt{10}d \end{pmatrix} = \begin{pmatrix} 5\sqrt{30} & 5\sqrt{10} \\ -5\sqrt{10} & 5\sqrt{30} \end{pmatrix}$$

$10\sqrt{10}a = 5\sqrt{30} \Rightarrow a = \frac{\sqrt{3}}{2}$

$10\sqrt{10}b = 5\sqrt{10} \Rightarrow b = \frac{1}{2}$

$10\sqrt{10}c = -5\sqrt{10} \Rightarrow c = -\frac{1}{2}$

$10\sqrt{10}d = 5\sqrt{30} \Rightarrow d = \frac{\sqrt{3}}{2}$

So $\mathbf{A} = \begin{pmatrix} 10\sqrt{10} & 0 \\ 0 & 10\sqrt{10} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$.

$\cos\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{30°, 330°\}$

$\sin\theta = -\frac{1}{2} \Rightarrow \theta = \{210°, 330°\}$

So $\mathbf{A}$ represents an enlargement, centred at the origin, by scale factor $10\sqrt{10}$, and a rotation, anticlockwise about the origin, by 330° (either order).

38 f)

$\det \mathbf{A} = (12 - 4\sqrt{3}) \times (12 - 4\sqrt{3}) - (12 + 4\sqrt{3}) \times (-12 - 4\sqrt{3})$

$$= (192 - 96\sqrt{3}) - (-192 - 96\sqrt{3}) = 384$$

$\sqrt{384} = 8\sqrt{6}$.

$$\begin{pmatrix} 8\sqrt{6} & 0 \\ 0 & 8\sqrt{6} \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 12 - 4\sqrt{3} & 12 + 4\sqrt{3} \\ -12 - 4\sqrt{3} & 12 - 4\sqrt{3} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 8\sqrt{6}a & 8\sqrt{6}b \\ 8\sqrt{6}c & 8\sqrt{6}d \end{pmatrix} = \begin{pmatrix} 12-4\sqrt{3} & 12+4\sqrt{3} \\ -12-4\sqrt{3} & 12-4\sqrt{3} \end{pmatrix}$$

$8\sqrt{6}a = 12-4\sqrt{3} \Rightarrow a = \frac{\sqrt{6}-\sqrt{2}}{4}$

$8\sqrt{6}b = 12+4\sqrt{3} \Rightarrow b = \frac{\sqrt{6}+\sqrt{2}}{4}$

$8\sqrt{6}c = -12-4\sqrt{3} \Rightarrow c = -\frac{\sqrt{6}+\sqrt{2}}{4}$

$8\sqrt{6}d = 12-4\sqrt{3} \Rightarrow d = \frac{\sqrt{6}-\sqrt{2}}{4}$

So $\mathbf{A} = \begin{pmatrix} 8\sqrt{6} & 0 \\ 0 & 8\sqrt{6} \end{pmatrix} \begin{pmatrix} \frac{\sqrt{6}-\sqrt{2}}{4} & \frac{\sqrt{6}+\sqrt{2}}{4} \\ -\frac{\sqrt{6}+\sqrt{2}}{4} & \frac{\sqrt{6}-\sqrt{2}}{4} \end{pmatrix}$.

$\cos\theta = \frac{\sqrt{6}-\sqrt{2}}{4} \Rightarrow \theta = \{75°, 285°\}$

$\sin\theta = -\frac{\sqrt{6}+\sqrt{2}}{4} \Rightarrow \theta = \{255°, 285°\}$

So **A** represents an enlargement, centred at the origin, by scale factor $8\sqrt{6}$, and a rotation, anti-clockwise about the origin, by 285° (either order).

39)

$$\mathbf{A} = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -k \\ -k & 0 \end{pmatrix}$$

$|\mathbf{A}| = -12 = 0 \times 0 - (-k) \times (-k) \Rightarrow k^2 = 12 \Rightarrow k = \sqrt{12} = 2\sqrt{3}.$

$\therefore \mathbf{A} = \begin{pmatrix} 0 & -2\sqrt{3} \\ -2\sqrt{3} & 0 \end{pmatrix}$.

40)

$$\begin{pmatrix} 5\sqrt{3} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 30° & -\sin 30° \\ \sin 30° & \cos 30° \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2\sqrt{2} \end{pmatrix} \begin{pmatrix} -\sqrt{3} & -4\sqrt{3} & -3\sqrt{3} \\ -\sqrt{2} & -\sqrt{2} & -6\sqrt{2} \end{pmatrix}$$

$$= \begin{pmatrix} 5\sqrt{3} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2\sqrt{2} \end{pmatrix} \begin{pmatrix} -\sqrt{3} & -4\sqrt{3} & -3\sqrt{3} \\ -\sqrt{2} & -\sqrt{2} & -6\sqrt{2} \end{pmatrix}$$

$$= \begin{pmatrix} 5\sqrt{3} & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & -4 & \frac{15}{2} \\ -\frac{5\sqrt{3}}{2} & -4\sqrt{3} & -\frac{27\sqrt{3}}{2} \end{pmatrix}$$

$$= \begin{pmatrix} \frac{5\sqrt{3}}{2} & -20\sqrt{3} & \frac{75\sqrt{3}}{2} \\ -\frac{5\sqrt{3}}{2} & -4\sqrt{3} & -\frac{27\sqrt{3}}{2} \end{pmatrix}$$

So the coordinates of the vertices of triangle X' are:
$(\frac{5\sqrt{3}}{2}, -\frac{5\sqrt{3}}{2})$, $(-20\sqrt{3}, -4\sqrt{3})$ and $\left(\frac{75\sqrt{3}}{2}, -\frac{27\sqrt{3}}{2}\right)$.

Area of X is

$$\frac{1}{2} \times (-\sqrt{3} - (-4\sqrt{3})) \times (-\sqrt{2} - (-6\sqrt{2})) = \frac{1}{2} \times 3\sqrt{3} \times 5\sqrt{2} = \frac{15\sqrt{6}}{2} \text{ units}^2$$

Area of X' is

$$\frac{15\sqrt{6}}{2} \times \begin{vmatrix} 5\sqrt{3} & 0 \\ 0 & 1 \end{vmatrix} \times \begin{vmatrix} 1 & 0 \\ 0 & 2\sqrt{2} \end{vmatrix} = \frac{15\sqrt{6}}{2} \times 5\sqrt{3} \times 2\sqrt{2} = 450 \text{ units}^2$$

41)

$$|\mathbf{A}| = \begin{vmatrix} 3 & -3\sqrt{3} \\ -3\sqrt{3} & -3 \end{vmatrix} = 3 \times (-3) - (-3\sqrt{3}) \times (-3\sqrt{3}) = -9 - 27 = -36$$

So $|\mathbf{R}| = 36$, $|\mathbf{P}| = 1$ (rotations maintain area and orientation), $|\mathbf{Q}| = -1$ (reflections maintain area but reverse orientation).

$\therefore \mathbf{R} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}$ so $\mathbf{R}$ represents an enlargement centred at the origin by scale factor 6.

$$\begin{pmatrix} \frac{1}{6} & 0 \\ 0 & \frac{1}{6} \end{pmatrix} \begin{pmatrix} 3 & -3\sqrt{3} \\ -3\sqrt{3} & -3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

So $\begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$ is a composition of a reflection and a rotation:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \quad \text{where } ad - bc = -1$$

$$\Rightarrow \begin{pmatrix} a\cos\theta + b\sin\theta & -a\sin\theta + b\cos\theta \\ c\cos\theta + d\sin\theta & -c\sin\theta + d\cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$a\cos\theta + b\sin\theta = \frac{1}{2} \quad (1)$

$-a\sin\theta + b\cos\theta = -\frac{\sqrt{3}}{2} \quad (2)$

$c\cos\theta + d\sin\theta = -\frac{\sqrt{3}}{2} \quad (3)$

$-c\sin\theta + d\cos\theta = -\frac{1}{2} \quad (4)$

(1) and (4):

$a\cos\theta + b\sin\theta = c\sin\theta - d\cos\theta$

$\Rightarrow b = c$ and $a = -d$.

(2) and (3):

$-a\sin\theta + b\cos\theta = c\cos\theta + d\sin\theta$

$\Rightarrow -a = d$ and $b = c$.

As $ad - bc = -1$, we must have $a(-a) - b(b) = -1 \Rightarrow -a^2 - b^2 = -1$

$\Rightarrow a^2 + b^2 = 1$.

Let $a = 1$ and $b = 0$, then $c = 0$ and $d = -1$.

$\therefore \mathbf{Q} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ so $\mathbf{Q}$ represents a reflection in the x-axis.

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \cos\theta & -\sin\theta \\ -\sin\theta & -\cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$\cos\theta = \frac{1}{2} \Rightarrow \theta = \{60°, 300°\}$

$\sin\theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \{60°, 120°\}$

$\therefore \mathbf{P} = \begin{pmatrix} \cos 60° & -\sin 60° \\ \sin 60° & \cos 60° \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$ so $\mathbf{P}$ represents a rotation by 60°, anticlockwise about the origin.

$\therefore \mathbf{A} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$.

Alternative Method:

Let $a = 0$ and $b = 1$, then $c = 1$ and $d = 0$.

$\therefore \mathbf{Q} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ so $\mathbf{Q}$ represents a reflection in the line $y = x$.

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \sin\theta & \cos\theta \\ \cos\theta & -\sin\theta \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

$\cos\theta = -\frac{\sqrt{3}}{2} \Rightarrow \theta = \{150°, 210°\}$

$\sin\theta = \frac{1}{2} \Rightarrow \theta = \{30°, 150°\}$

$\therefore \mathbf{P} = \begin{pmatrix} \cos 150° & -\sin 150° \\ \sin 150° & \cos 150° \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$ so $\mathbf{P}$ represents a rotation by 150°, anticlockwise about the origin.

$\therefore \mathbf{A} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$.

So $\mathbf{A} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}$ or $\mathbf{A} = \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} -\frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & -\frac{\sqrt{3}}{2} \end{pmatrix}$.

42)

Let $\mathbf{A} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$.

Consider **AB**:

$$\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{5} \\ 5 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1+ab & a \\ b & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{5} \\ 5 & 1 \end{pmatrix}$$

$a = -\frac{1}{5}$ and $b = 5$.

Check: $1 + ab = 1 + (-\frac{1}{5}) \times 5 = 0$.

If we had instead considered **BA**:

$$\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix} \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{5} \\ 5 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & a \\ b & ab+1 \end{pmatrix} = \begin{pmatrix} 0 & -\frac{1}{5} \\ 5 & 1 \end{pmatrix}$$

We would then have had $1 = 0$, so **BA** is not the correct order.

So $\mathbf{A} = \begin{pmatrix} 1 & -\frac{1}{5} \\ 0 & 1 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$.

43 a)

Shear, with the x-axis fixed, with $(0, 1)$ mapped to $(-5, 1)$.

43 b)

Let $\mathbf{C} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

$$\mathbf{CA} = \mathbf{B} \Rightarrow \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & -5 \\ 0 & 1 \end{pmatrix} = \tfrac{1}{2} \begin{pmatrix} \sqrt{2} & -6\sqrt{2} \\ \sqrt{2} & -4\sqrt{2} \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a & b-5a \\ c & d-5c \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -3\sqrt{2} \\ \frac{\sqrt{2}}{2} & -2\sqrt{2} \end{pmatrix}$$

$a = \frac{\sqrt{2}}{2}$ and $c = \frac{\sqrt{2}}{2}$

$b - 5 \times \frac{\sqrt{2}}{2} = -3\sqrt{2} \Rightarrow b = -\frac{\sqrt{2}}{2}$

$d - 5 \times \frac{\sqrt{2}}{2} = -2\sqrt{2} \Rightarrow d = \frac{\sqrt{2}}{2}$

$\therefore \mathbf{C} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix}$.

This is similar to the general rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

$\cos\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 315°\}$

$\sin\theta = \frac{\sqrt{2}}{2} \Rightarrow \theta = \{45°, 135°\}$

So it is a rotation by 45°, anticlockwise about the origin.

44)

Let $\mathbf{A} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix}$.

$$\mathbf{ABC} = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & c \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & c \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix} \Rightarrow \begin{pmatrix} a & -1+ac \\ 1 & c \end{pmatrix} = \begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}$$

$\therefore a = 2$ and $c = 4$.

Check: $-1 + ac = -1 + 2 \times 4 = 7$.

So $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\mathbf{C} = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$.

45)

Area of triangle $X = \frac{1}{2} \times 3 \times 2 = 3$ units^2.

Area of triangle $X' = \frac{1}{2} \times 3 \times 4 = 6$ units^2.

$\frac{6}{3} = 2$ so the matrix that represents the stretch is either $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ or $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$.

As $B(3, 0)$ is mapped to $B'(0, 3)$, this can be achieved via a rotation, anticlockwise about the origin, by 90°, followed by a stretch parallel to the x-axis, scale factor 2, followed by a shear that fixes the y-axis. The rotation matrix would be $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. If the stretch is parallel to the x-axis then we have $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.

So far we have:

$$\mathbf{M} = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Then:

$$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 3 & 3 \\ 2 & 0 & 2 \end{pmatrix} = \begin{pmatrix} -4 & 0 & -4 \\ 12 & 3 & 15 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 0 & -2 \\ 0 & 3 & 3 \end{pmatrix} = \begin{pmatrix} -4 & 0 & -4 \\ 12 & 3 & 15 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix} \begin{pmatrix} -4 & 0 & -4 \\ 0 & 3 & 3 \end{pmatrix} = \begin{pmatrix} -4 & 0 & -4 \\ 12 & 3 & 15 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -4 & 0 & -4 \\ -4k & 3 & -4k+3 \end{pmatrix} = \begin{pmatrix} -4 & 0 & -4 \\ 12 & 3 & 15 \end{pmatrix}$$

$-4k = 12 \Rightarrow k = -3$

$-4k + 3 = 15 \Rightarrow k = -3$

$\therefore \mathbf{M} = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.